C Probability Review

In this appendix, we give a brief review of some basic notions of probability and will also define the notation that is used throughout the textbook.

C.1 Probability

A *probability space* is a tuple consisting of three components: a *sample space*, an *events set*, and a *probability distribution*:

- *sample space* Ω: Ω is the set of all elementary events or outcomes possible in a trial, for example, each of the six outcomes in $\{1, \ldots, 6\}$ when casting a die.

- *events set* $\mathcal{F}$: $\mathcal{F}$ is a *σ-algebra*, that is a set of subsets of Ω containing Ω that is closed under complementation and countable union (therefore also countable intersection). An example of an event may be "the die lands on an odd number".

- *probability distribution*: $\mathbb{P}$ is a mapping from the set of all events $\mathcal{F}$ to $[0, 1]$ such that $\mathbb{P}[\Omega] = 1$, $\mathbb{P}[\emptyset] = 1$, and, for all mutually exclusive events $A_1, \ldots, A_n$,
$$\mathbb{P}[A_1 \cup \ldots \cup A_n] = \sum_{i=1}^{n} \mathbb{P}[A_i].$$
The discrete probability distribution associated with a fair die can be defined by $\mathbb{P}[A_i] = 1/6$ for $i \in \{1 \ldots 6\}$, where A_i is the event that the die lands on value i.

C.2 Random variables

Definition C.1 (Random variables) *A* random variable X *is a function* $X : \Omega \to \mathbb{R}$ *that is* measurable*, that is such that for any interval I, the subset of the sample space $\{\omega \in \Omega \colon X(\omega) \in I\}$ is an event.*

The *probability mass function* of a discrete random variable X is defined as the function $x \mapsto \mathbb{P}[X = x]$. The *joint probability mass function* of discrete random variables X and Y is defined as the function $(x, y) \mapsto \mathbb{P}[X = x \wedge Y = y]$.

A probability distribution is said to be *absolutely continuous* when it admits a *probability density function*, that is a function f associated to a real-valued random variable X that satisfies for all $a, b \in \mathbb{R}$
$$\mathbb{P}[a \leq X \leq b] = \int_a^b f(x)dx\,. \tag{C.1}$$

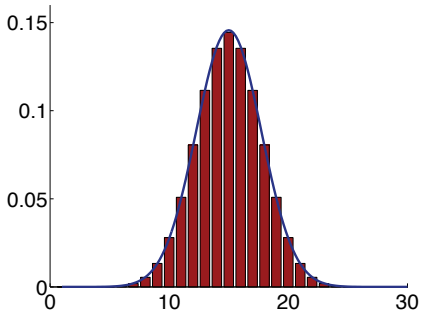

Figure C.1
Approximation of the binomial distribution (in red) by a normal distribution (in blue).

Definition C.2 (Binomial distribution) *A random variable X is said to follow a* binomial distribution $B(n,p)$ *with $n \in \mathbb{N}$ and $p \in [0,1]$ if for any $k \in \{0,1,\ldots,n\}$,*

$$\mathbb{P}[X = k] = \binom{n}{k} p^k (1-p)^{n-k} .$$

Definition C.3 (Normal distribution) *A random variable X is said to follow a* normal *(or* Gaussian*)* distribution $N(\mu, \sigma^2)$ *with $\mu \in \mathbb{R}$ and $\sigma > 0$ if its probability density function is given by,*

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) .$$

The standard normal *distribution $N(0,1)$ is the normal distribution with zero mean and unit variance.*

The normal distribution is often used to approximate a binomial distribution. Figure C.1 illustrates that approximation.

Definition C.4 (Laplace distribution) *A random variable X is said to follow a* Laplace distribution *with location parameter $\mu \in \mathbb{R}$ and scale parameter $b > 0$ if its probability density function is given by,*

$$f(x) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right) .$$

Definition C.5 (Gibbs distributions) *Given a set $\mathcal{X}$ and* feature function $\Phi \colon \mathcal{X} \to \mathbb{R}^N$*, a random variable X is said to follow a* Gibbs distribution *with parameter $\mathbf{w} \in \mathbb{R}^N$ if for any $x \in \mathcal{X}$,*

$$\mathbb{P}[X = x] = \frac{\exp(\mathbf{w} \cdot \Phi(x))}{\sum_{x \in \mathcal{X}} \exp(\mathbf{w} \cdot \Phi(x))} .$$

The normalizing quantity in the denominator $Z = \sum_{x \in \mathcal{X}} \exp(\mathbf{w} \cdot \Phi(x))$ is also called the partition function*.*

Definition C.6 (Poisson distribution) *A random variable X is said to follow a* Poisson distribution *with $\lambda > 0$ if for any $k \in \mathbb{N}$,*

$$\mathbb{P}[X = k] = \frac{\lambda^k e^{-\lambda}}{k!} .$$

The definition of the following family of distributions uses the notion of independence of random variables defined in the next section.

Definition C.7 (χ^2 distribution) *The χ^2* distribution *(or* chi-squared distribution*) with k* degrees of freedom *is the distribution of the sum of the squares of k independent random variables, each following a standard normal distribution.*

C.3 Conditional probability and independence

Definition C.8 (Conditional probability) *The* conditional probability *of event A given event B is defined by*

$$\mathbb{P}[A \mid B] = \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}, \tag{C.2}$$

when $\mathbb{P}[B] \neq 0$.

Definition C.9 (Independence) *Two events A and B are said to be* independent *if*

$$\mathbb{P}[A \cap B] = \mathbb{P}[A]\,\mathbb{P}[B]. \tag{C.3}$$

Equivalently, A and B are independent iff $\mathbb{P}[A \mid B] = \mathbb{P}[A]$ when $\mathbb{P}[B] \neq 0$.

A sequence of random variables is said to be *independent and identically distributed (i.i.d.)* when the random variables are mutually independent and follow the same distribution.

The following are basic probability formulae related to the notion of conditional probability. They hold for any events A, B, and $A_1, \ldots, A_n$, with the additional constraint $\mathbb{P}[B] \neq 0$ needed for the Bayes formula to be well defined:

$$\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B] \qquad (\textit{sum rule}) \tag{C.4}$$

$$\mathbb{P}[\bigcup_{i=1}^{n} A_i] \leq \sum_{i=1}^{n} \mathbb{P}[A_i] \qquad (\textit{union bound}) \tag{C.5}$$

$$\mathbb{P}[A \mid B] = \frac{\mathbb{P}[B \mid A]\,\mathbb{P}[A]}{\mathbb{P}[B]} \qquad (\textit{Bayes formula}) \tag{C.6}$$

$$\mathbb{P}[\bigcap_{i=1}^{n} A_i] = \mathbb{P}[A_1]\,\mathbb{P}[A_2 \mid A_1] \cdots \mathbb{P}[A_n \mid \bigcap_{i=1}^{n-1} A_i] \qquad (\textit{chain rule}). \tag{C.7}$$

The sum rule follows immediately from the decomposition of $A \cup B$ as the union of the disjoint sets A and $(B - A \cap B)$. The union bound is a direct consequence of the sum rule. The Bayes formula follows immediately from the definition of conditional probability and the observation that: $\mathbb{P}[A|B]\,\mathbb{P}[B] = \mathbb{P}[B|A]\,\mathbb{P}[A] = \mathbb{P}[A \cap B]$. Similarly, the chain rule follows the observation that $\mathbb{P}[A_1]\,\mathbb{P}[A_2|A_1] = \mathbb{P}[A_1 \cap A_2]$; using the same argument shows recursively that the product of the first k terms of the right-hand side equals $\mathbb{P}[\bigcap_{i=1}^{k} A_i]$.

Finally, assume that $\Omega = A_1 \cup A_2 \cup \ldots \cup A_n$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, i.e., the A_is are mutually disjoint. Then, the following formula is valid for any event B:

$$\mathbb{P}[B] = \sum_{i=1}^{n} \mathbb{P}[B \mid A_i]\,\mathbb{P}[A_i] \qquad (\textit{theorem of total probability}). \tag{C.8}$$

This follows the observation that $\mathbb{P}[B \mid A_i]\,\mathbb{P}[A_i] = \mathbb{P}[B \cap A_i]$ by definition of the conditional probability and the fact that the events $B \cap A_i$ are mutually disjoint.

C.4 Expectation and Markov's inequality

Definition C.10 (Expectation) *The* expectation *or* mean *of a random variable X is denoted by $\mathbb{E}[X]$ and defined by*

$$\mathbb{E}[X] = \sum_{x} x\,\mathbb{P}[X = x]. \tag{C.9}$$

When X follows a probability distribution $\mathcal{D}$, we will also write $\mathbb{E}_{x \sim \mathcal{D}}[x]$ instead of $\mathbb{E}[X]$ to explicitly indicate the distribution. A fundamental property of expectation, which is straightforward to verify using its definition, is that it is linear, that is, for any two random variables X and Y and any $a, b \in \mathbb{R}$, the following holds:

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]. \tag{C.10}$$

Furthermore, when X and Y are independent random variables, then the following identity holds:

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]. \tag{C.11}$$

Indeed, by definition of expectation and of independence, we can write

$$\begin{aligned}\mathbb{E}[XY] = \sum_{x,y} xy\,\mathbb{P}[X = x \wedge Y = y] &= \sum_{x,y} xy\,\mathbb{P}[X = x]\,\mathbb{P}[Y = y] \\ &= \Big(\sum_x x\,\mathbb{P}[X = x]\Big)\Big(\sum_y y\,\mathbb{P}[Y = y]\Big),\end{aligned}$$

where in the last step we used Fubini's theorem . The following provides a simple bound for a non-negative random variable in terms of its expectation, known as *Markov's inequality*.

Theorem C.11 (Markov's inequality) *Let X be a non-negative random variable with $\mathbb{E}[X] < \infty$. Then for all $t > 0$,*

$$\mathbb{P}\left[X \geq t\,\mathbb{E}[X]\right] \leq \frac{1}{t}. \tag{C.12}$$

Proof: The proof steps are as follows:

$$\begin{aligned}\mathbb{P}[X \geq t\,\mathbb{E}[X]] &= \sum_{x \geq t\,\mathbb{E}[X]} \mathbb{P}[X = x] && \text{(by definition)} \\ &\leq \sum_{x \geq t\,\mathbb{E}[X]} \mathbb{P}[X = x]\frac{x}{t\,\mathbb{E}[X]} && \left(\text{using } \frac{x}{t\,\mathbb{E}[X]} \geq 1\right) \\ &\leq \sum_x \mathbb{P}[X = x]\frac{x}{t\,\mathbb{E}[X]} && \text{(extending non-negative sum)} \\ &= \mathbb{E}\left[\frac{X}{t\,\mathbb{E}[X]}\right] = \frac{1}{t} && \text{(linearity of expectation).}\end{aligned}$$

This concludes the proof. □

C.5 Variance and Chebyshev's inequality

Definition C.12 (Variance — Standard deviation) *The* variance *of a random variable X is denoted by* $\mathrm{Var}[X]$ *and defined by*

$$\mathrm{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]. \tag{C.13}$$

The standard deviation *of a random variable X is denoted by σ_X and defined by*

$$\sigma_X = \sqrt{\mathrm{Var}[X]}. \tag{C.14}$$

For any random variable X and any $a \in \mathbb{R}$, the following basic properties hold for the variance, which can be proven straightforwardly:

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 \tag{C.15}$$

$$\mathrm{Var}[aX] = a^2\,\mathrm{Var}[X]. \tag{C.16}$$

Furthermore, when X and Y are independent, then

$$\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]. \tag{C.17}$$

Indeed, using the linearity of expectation and the identity $\mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[XY] = 0$ which holds by the independence of X and Y, we can write

$$\begin{aligned}\mathrm{Var}[X + Y] &= \mathbb{E}[(X + Y)^2] - \mathbb{E}[X + Y]^2 \\ &= \mathbb{E}[X^2 + Y^2 + 2XY] - (\mathbb{E}[X]^2 + \mathbb{E}[Y]^2 + 2\,\mathbb{E}[XY]) \\ &= (\mathbb{E}[X^2] - \mathbb{E}[X]^2) + (\mathbb{E}[Y^2] - \mathbb{E}[Y]^2) + 2(\mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[XY]) \\ &= \mathrm{Var}[X] + \mathrm{Var}[Y].\end{aligned}$$

The following inequality known as *Chebyshev's inequality* bounds the deviation of a random variable from its expectation in terms of its standard deviation.

Theorem C.13 (Chebyshev's inequality) *Let X be a random variable with* $\mathrm{Var}[X] < +\infty$. *Then, for all $t > 0$, the following inequality holds:*

$$\mathbb{P}\left[|X - \mathbb{E}[X]| \geq t\sigma_X\right] \leq \frac{1}{t^2}. \tag{C.18}$$

Proof: Observe that:

$$\mathbb{P}\left[|X - \mathbb{E}[X]| \geq t\sigma_X\right] = \mathbb{P}[(X - \mathbb{E}[X])^2 \geq t^2\sigma_X^2].$$

The result follows by application of Markov's inequality to $(X - \mathbb{E}[X])^2$. □

We will use Chebyshev's inequality to prove the following theorem.

Theorem C.14 (Weak law of large numbers) *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables with the same mean μ and variance $\sigma^2 < \infty$. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$, then, for any $\epsilon > 0$,*

$$\lim_{n\to\infty} \mathbb{P}[|\overline{X}_n - \mu| \geq \epsilon] = 0. \tag{C.19}$$

Proof: Since the variables are independent, we can write

$$\mathrm{Var}[\overline{X}_n] = \sum_{i=1}^n \mathrm{Var}\left[\frac{X_i}{n}\right] = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Thus, by Chebyshev's inequality (with $t = \epsilon/(\mathrm{Var}[\overline{X}_n])^{1/2}$), the following holds:

$$\mathbb{P}[|\overline{X}_n - \mu| \geq \epsilon] \leq \frac{\sigma^2}{n\epsilon^2},$$

which implies (C.19). □

Example C.15 (Applying Chebyshev's inequality) Suppose we roll a pair of fair dice n times. Can we give a good estimate of the total value of the n rolls? If we compute the mean and variance, we find $\mu = 7n$ and $\sigma^2 = 35/6n$ (we leave it to the reader to verify these expressions). Thus, applying Chebyshev's inequality, we see that the final sum will lie within $7n \pm 10\sqrt{\frac{35}{6}n}$ in at least 99 percent of all experiments. Therefore, the odds are better than 99 to 1 that the sum will be between 6.975M and 7.025M after 1M rolls.

Definition C.16 (Covariance) *The* covariance *of two random variables X and Y is denoted by* $\mathrm{Cov}(X, Y)$ *and defined by*

$$\mathrm{Cov}(X, Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right]. \tag{C.20}$$

Two random variables X and Y are said to be *uncorrelated* when $\mathrm{Cov}(X, Y) = 0$. It is straightforward to see that if two random variables X and Y are independent then they are uncorrelated, but the converse does not hold in general. The covariance defines a positive semidefinite and symmetric bilinear form:

- symmetry: $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$ for any two random variables X and Y;
- bilinearity: $\mathrm{Cov}(X + X', Y) = \mathrm{Cov}(X, Y) + \mathrm{Cov}(X', Y)$ and $\mathrm{Cov}(aX, Y) = a\,\mathrm{Cov}(X, Y)$ for any random variables X, X', and Y and $a \in \mathbb{R}$;
- positive semidefiniteness: $\mathrm{Cov}(X, X) = \mathrm{Var}[X] \geq 0$ for any random variable X.

The following Cauchy-Schwarz inequality holds for random variables X and Y with $\mathrm{Var}[X] < +\infty$ and $\mathrm{Var}[Y] < +\infty$:

$$|\,\mathrm{Cov}(X, Y)| \leq \sqrt{\mathrm{Var}[X]\,\mathrm{Var}[Y]}. \tag{C.21}$$

The following definition extends the notion of covariance to a vector of random variables.

Definition C.17 *The* covariance matrix *of a vector of random variables* $\mathbf{X} = (X_1, \ldots, X_N)$ *is the matrix in* $\mathbb{R}^{N\times N}$ *denoted by* $\mathbf{C}(\mathbf{X})$ *and defined by*

$$\mathbf{C}(\mathbf{X}) = \mathbb{E}\left[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top\right]. \tag{C.22}$$

Thus, $\mathbf{C}(\mathbf{X}) = (\mathrm{Cov}(X_i, X_j))_{ij}$. It is straightforward to show that

$$\mathbf{C}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}[\mathbf{X}]\,\mathbb{E}[\mathbf{X}]^\top. \tag{C.23}$$

We close this appendix with the following well-known theorem of probability.

Theorem C.18 (Central limit theorem) *Let* $X_1, \ldots, X_n$ *be a sequence of i.i.d. random variables with mean* μ *and standard deviation* σ*. Let* $\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ *and* $\overline{\sigma}_n^2 = \sigma^2/n$*. Then,* $(\overline{X}_n - \mu)/\overline{\sigma}_n$ converges to the $N(0,1)$ in distribution*, that is for any* $t \in \mathbb{R}$*,*

$$\lim_{n\to\infty} \mathbb{P}[(\overline{X}_n - \mu)/\overline{\sigma}_n \le t] = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx\,.$$

C.6 Moment-generating functions

The expectation $\mathbb{E}[X^p]$ is called the *pth-moment* of the random variable X. The moment-generating function of a random variable X is a key function from which its different moments can be straightforwardly computed via differentiation at zero. It can therefore be crucial for specifying the distribution of X or analyzing its properties.

Definition C.19 (Moment-generating function) *The* moment-generating function *of a random variable* X *is the function* $M_X : t \mapsto \mathbb{E}[e^{tX}]$ *defined over the set of* $t \in \mathbb{R}$ *for which the expectation is finite.*

If M_X is differentiable at zero, then the pth-moment of X is given by $\mathbb{E}[X^p] = M_X^{(p)}(0)$. We will present in the next chapter a general bound on the moment-generating function of a zero-mean bounded random variable (Lemma D.1). Here, we illustrate its computation in two special cases.

Example C.20 (Standard normal distribution) Let X be a random variable following a normal distribution with mean 0 and variance 1. Then, M_X is defined for all $t \in \mathbb{R}$ by

$$M_X(t) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} e^{tx} dx = e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx = e^{\frac{t^2}{2}}\,, \tag{C.24}$$

by recognizing that the last integrand is the probability density function of a normal distribution with mean t and variance 1.

Example C.21 (χ^2 distribution) Let X be a random variable following a χ^2 distribution with k degrees of freedom. We can write $X = \sum_{i=1}^k X_i^2$ where the X_is are independent and follow a standard normal distribution.

Let $t < 1/2$. By the i.i.d. assumption about the variables X_i, we can write

$$\mathbb{E}[e^{tX}] = \mathbb{E}\left[\prod_{i=1}^k e^{tX_i^2}\right] = \prod_{i=1}^k \mathbb{E}\left[e^{tX_i^2}\right] = \mathbb{E}\left[e^{tX_1^2}\right]^k.$$

By definition of the standard normal distribution, we have

$$\begin{aligned}\mathbb{E}[e^{tX_1^2}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{tx^2} e^{\frac{-x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{(1-2t)\frac{-x^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{e^{\frac{-u^2}{2}}}{\sqrt{1-2t}} du = (1-2t)^{-\frac{1}{2}}\,,\end{aligned}$$

where we used the change of variable $u = \sqrt{1-2t}\,x$. In view of that, the moment-generating function of the χ^2 distribution is given by

$$\forall t < 1/2,\ M_X(t) = \mathbb{E}[e^{tX}] = (1-2t)^{-\frac{k}{2}}\,. \tag{C.25}$$

C.7 Exercises

C.1 Let $f\colon (0,+\infty) \to \mathbb{R}_+$ be a function admitting an inverse f^{-1} and let X be a random variable. Show that if for any $t > 0$, $\mathbb{P}[X > t] \leq f(t)$, then, for any $\delta > 0$, with probability at least $1-\delta$, $X \leq f^{-1}(\delta)$.

C.2 Let X be a discrete random variable taking non-negative integer values. Show that $\mathbb{E}[X] = \sum_{n\geq 1} \mathbb{P}[X \geq n]$ (*Hint*: rewrite $\mathbb{P}[X = n]$ as $\mathbb{P}[X \geq n] - \mathbb{P}[X \geq n+1]$).